their injuries. After the tests you just performed, you are in the driver's seat, and how you use your body throughout the day will determine whether or not you have pain!

Stuart McGill once told me, "During recovery, what you *don't* do is often just as important as what you *do* do." Remember this motto as you begin your journey to eliminate your pain and return to the activities you love. Below, I have laid out some suggestions based on the movement diagnosis that you may fall under. These are general guidelines and certainly not hard rules.

Flexion Intolerance

SIGNS & SYMPTOMS:

Your pain tends to come out when your spine is in a flexed position or moves into flexion.

GUIDELINES:

In bed: Don't sit straight up when getting out of bed. Roll to your side and then push up with your arms. If lying on your stomach feels good, remain on your stomach for a few minutes two or three times a day.

Sitting: Place a small rolled towel under your low back to keep it from rounding. Sit up tall and don't slouch!

Standing: Make sure you're standing tall and not slouching throughout the day.

Picking things up: Instead of bending over to grab something (like clothes out of a laundry basket), kneel down to keep your back from rounding. Also learn how to hinge at the hips while maintaining a neutral spine. This will keep your low back from entering into too much flexion when picking objects off the ground.

Extension Intolerance

SIGNS & SYMPTOMS:

Your pain tends to come out when your spine is in an overarched position or moves into extension.

GUIDELINES:

In bed: If you're a stomach sleeper, place a pillow under your belly. If you sleep on your back, place a pillow under your knees. Both will help decrease the amount of extension in your low back to make sleeping more comfortable.

Sitting: Relax your back against the back of a chair. Don't sit on the edge, which may cause you to overarch your spine.

Standing: Check your posture in the mirror (lateral view). Make sure your low back is in neutral and not excessively arched.

Picking things up: Think about moving from your hips more than your low back (emphasize hip extension over low back extension). Kneeling or squatting down to pick something off the ground is also helpful.

Rotation with Extension Intolerance

SIGNS & SYMPTOMS:

Your low back becomes painful when extension is coupled with a rotational or twisting movement.

GUIDELINES:

In bed: If you're a side sleeper, place a pillow between your knees to limit excessive rotation at your spine and hips. When getting out of bed, roll your legs and trunk together.

Sitting: Try not to cross your legs. Also avoid leaning or shifting your weight excessively to one hip.

Standing: Stand with normal posture, but make sure you are placing equal weight through both legs. Having one foot forward and one foot back may cause unwanted rotation at the hip/spine.

Picking things up: Limit excessive rotation in your trunk. Move with your hips/legs, not with your back.

Load Intolerance (Dynamic or Compressive)

SIGNS & SYMPTOMS:

Your pain comes out when lifting something heavy or when performing physical tasks that load your spine rapidly, such as performing Olympic lifts or running.

GUIDELINES:

- Avoid running or any other task that places a load or shock through your body.
- Limit your lifting in the gym to pain-free tasks at this time! Don't be a tough guy and push through pain; it will only stretch out the time it takes to become pain-free.

Can I Still Lift?

During these next few weeks and possibly months, you must consider limiting your heavy training and exercising only with movements that do not create pain. Pain not only is an annoyance but also changes the way you move. [63]

This means the back pain you are experiencing is hindering your ability to use good technique when lifting. Not only does this have direct implications on your performance, but training through pain and loading your body with compromised technique can worsen the injury.

I have worked with many athletes who have continued to push through heavy programming while dealing with back pain. It does not end well. If you want to eliminate your pain and return to high-performance lifting, you must tone down any lifting that creates pain at this time. This doesn't mean you should stop working out! You must change how you are lifting in the short term to allow healing to take place.

When Is It Time to See a Doctor?

If none of these strategies works, or if your pain is progressively getting worse, please seek out a skilled clinician (physical therapist, chiropractor, or orthopedic doctor who specializes in back pain). If you are experiencing a recent unintentional loss in weight, incontinence, pain, or numbness in your abdomen or pelvic floor, I highly recommend you make an appointment to see your medical doctor.

The Rebuilding Process

By now, you should have a good idea of which specific triggers cause your pain. You should also have an understanding of which postures and movements you can perform pain-free. It is time to start addressing a common weak link in almost all cases of back pain: core instability.

Think about the last time you went to your family doctor with a complaint of pain. There's a good chance the doc uttered these words: "I recommend not lifting for a few weeks." Sound familiar? Many people find some short-term relief of their symptoms by following these orders. It makes sense on paper. If deadlifting causes your back to hurt, not deadlifting will likely decrease your pain! Problem solved, right?

Wrong.

Chances are the pain will eventually return because you never addressed *why* the problem started in the first place.

Eliminating the movement, posture, or load that causes your pain is only half the battle in fixing any injury. Anyone can tell you to stop doing something that hurts. Eliminating symptoms and building your body to become more resilient to injury requires a different and more active approach.

To kick-start this active approach, let's learn about the core and how it relates to both the cause and the fix for low back injuries.

Core Stability

Imagine a symphony orchestra. Just as every musician in an orchestra must play their instrument in a united manner despite constant changes in tempo and volume, your body must coordinate every muscle to create purposeful and sound movement.

The muscles that surround the spine—the front and side abdominal muscles, the erector muscles of the back, and even the larger muscles that span multiple joints, like the lats and psoas muscles—are considered the "core" of the body. It may surprise you that the glutes are also an important part of the core (something you'll learn about very soon!). These muscles must work together to enhance the stability of the spine.

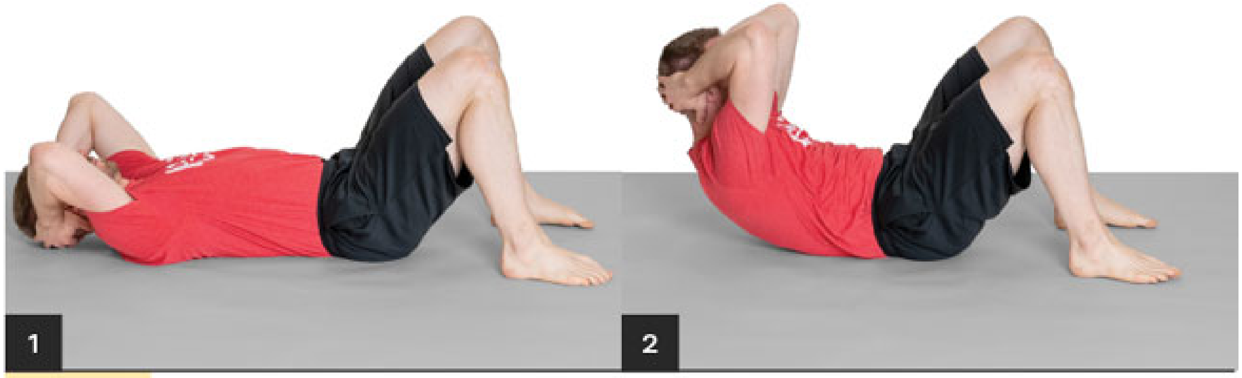
Spinal stability is something Dr. McGill has been able to define and measure in his work. First, when muscles contract, they create force and stiffness. It is the stiffness that is important for stability. Think of the spine as a flexible rod that needs to be stiffened to bear load. This is the role of the muscles. Through his research, he has measured athletes who failed to obtain appropriate muscular stiffness around the spine by coordinating muscle activation and their subsequent injuries and pain.

Second, the body functions as a linked system, and distal movement requires proximal stability. Consider trying to move a finger back and forth very quickly—the wrist needs to be stable; otherwise, the entire hand would move. Now, using the same principle, consider the action of walking. The pelvis must be stable in relation to the spine; otherwise, the left hip would drop as the left leg swung forward to take a step. This core stiffness is non-negotiable to enable walking. Thus all body movement needs appropriate coordination of muscles. To move, run, or squat requires spinal stiffness and core stability.

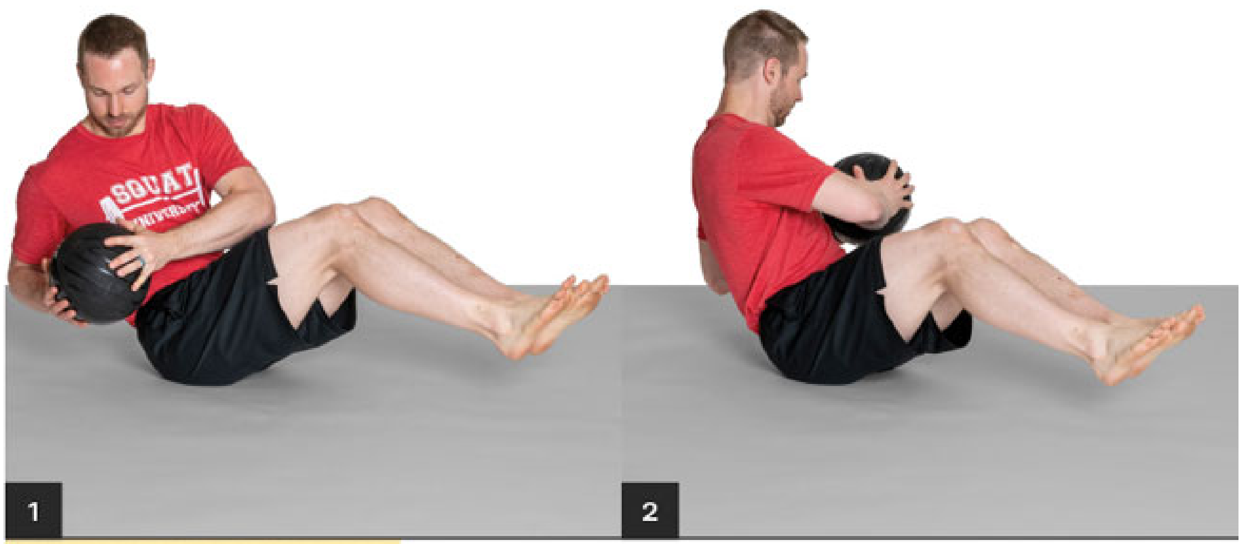
When the core fails to meet the stability demands placed on the body during certain lifts, parts of the spine are overloaded with forces that increase injury risk, and performance suffers—much like a bad musician who is playing off-pitch or off-beat, which would quickly affect the rest of the orchestra. Each muscle that surrounds or attaches to the spine must play in concert with others to produce safe, efficient and, when needed, powerful movement.

Where Do We Start?

There are two general approaches taken to address the core. The first and more common method (which you'll see in fitness clubs across the world) is dynamic strengthening exercises such as crunches, back extensions, and Russian twists. Traditionally, coaches and medical practitioners have used these exercises that build strength through movement with the mindset that a stronger core will give the spine less chance to buckle and break under tension.



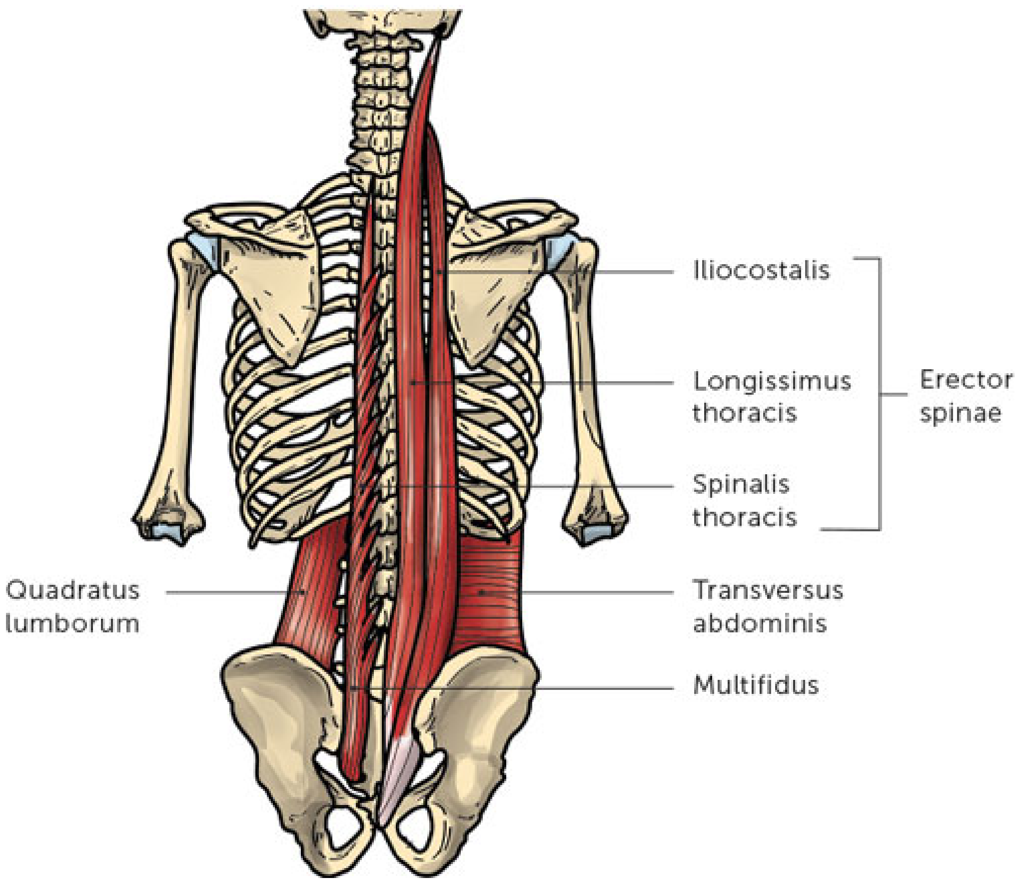
CRUNCH



RUSSIAN TWIST WITH BALL

RADIO TOWER WITH GUY-WIRES

ANATOMY OF CORE

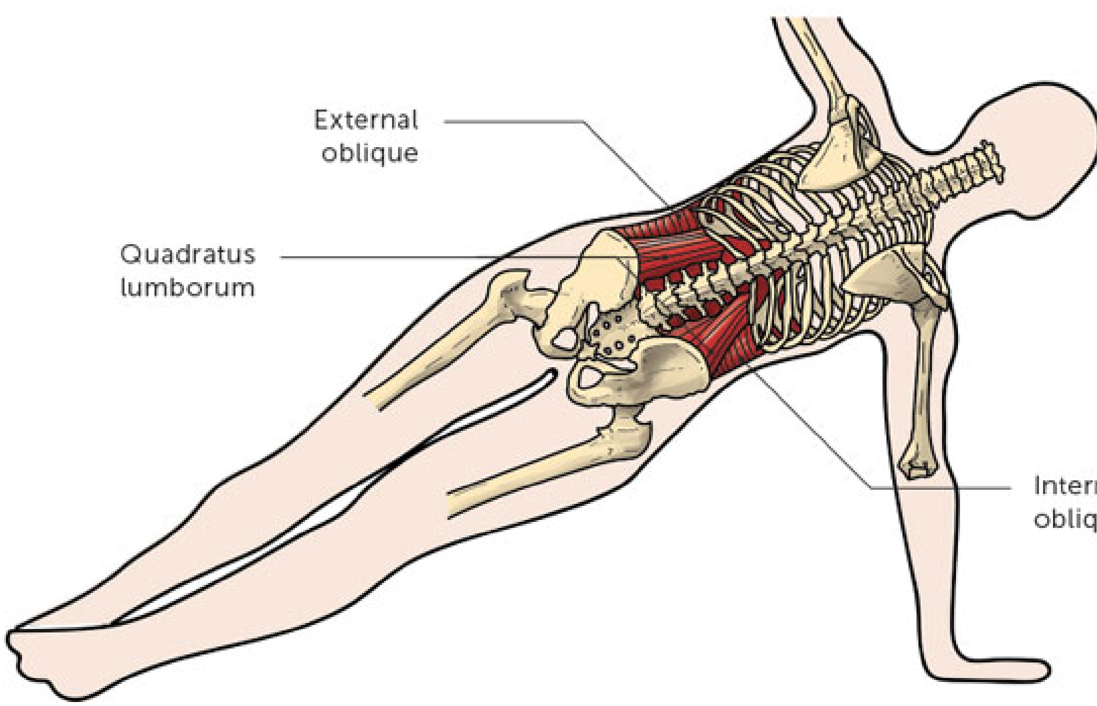


However, here's what most people don't understand. Many people who develop back pain already have strong backs! [64] While exercises like Russian twists, sit-ups, and back extensions from a GHD machine may be great at increasing strength, they do little to increase core stiffness. [65]

To enhance the quality of stiffness, you must train the core differently. This comes through the second approach of using isometric exercises built to enhance muscular endurance and coordination.

An isometric describes when a muscle or group of muscles are activated and contracted, but there is no change in the joints they cross. For example, during a side plank, the lateral oblique and quadratus lumborum (QL) muscles are very active, yet the spine and hips remain still and do not move. Research has found that isometric exercises to enhance muscular endurance are far superior when compared to dynamic strengthening exercises in enhancing spinal stiffness and stability (making them ideal not only for rehabilitation of back injuries but also in the training and enhancement of athletic performance). [66]

QL IN SIDE PLANK



Have you ever pushed your body to the max while performing a high-volume squat session and ended up failing to complete the set? I have many times during my lifting career. You're tired, and while trying to grind out three more reps, you come up one short and have to drop the bar to the ground. Sound familiar? If this is something you've experienced, go back and review that specific set, but instead of analyzing the failed last rep, check out the one right before it. Often, the "smoking gun" is a drop in core stability—you became a little loose in your ascent, your spine rounded ever so slightly, or your hips shifted a bit to the side.

As soon as your brain senses a loss in core stability, it shuts down "neural drive" to your muscles, making it much more difficult to squeeze out another rep. Without the core functioning correctly, your body has two choices. The first is to allow technique to falter even more as you grind out the last rep (allowing for even more spinal rounding or hip shifting). While you may "get the job done," your risk for injury dramatically increases. The second option: fail the rep, drop the barbell to the ground, and live to squat another day.

You see, your brain shuts down how hard your body can push through that next repetition as a protective mechanism. The way to train your body to withstand this breakdown and maintain a high "neural drive" to your muscles is to address the failure mechanism head-on (the *why* behind the collapse): core stability endurance.

The definition of stability is the ability to limit excessive or unwanted motion. Therefore, the traditional way the fitness and rehab world has approached addressing the core for years has been completely backward! This is why someone can have a ripped six-pack and yet easily experience technique breakdown when it comes to deadlifting or performing a squat.

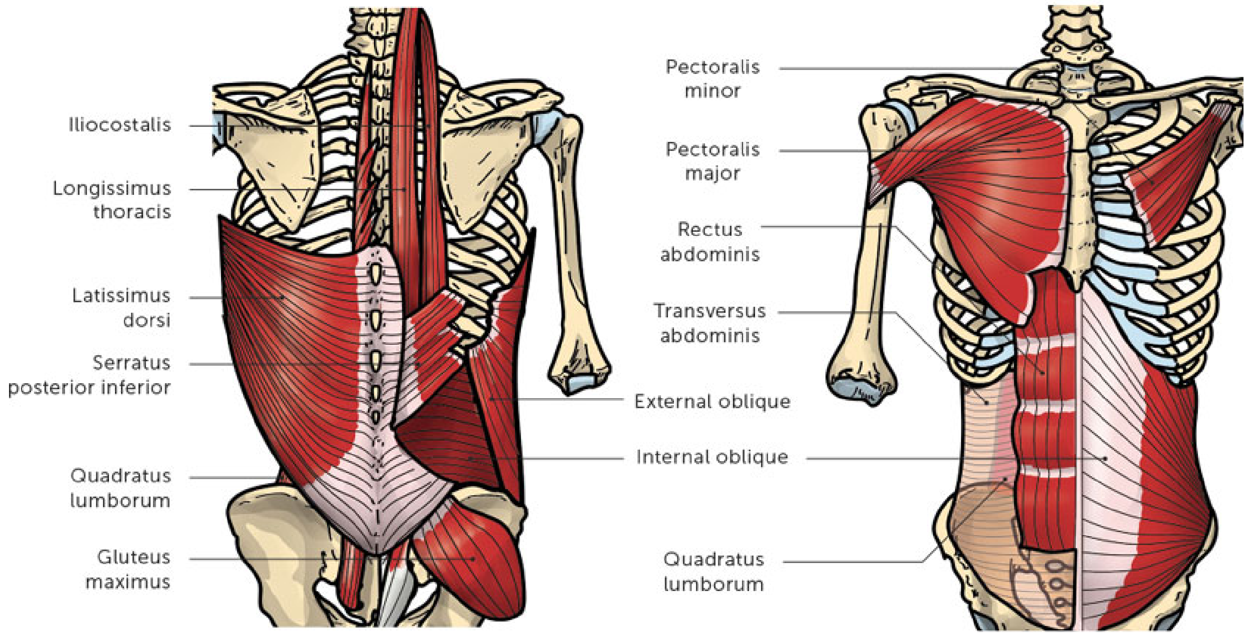
To stiffen the torso and limit excessive motion, every muscle of the core must co-contract or work together. **When this is done correctly through a bracing action, your body creates its natural "weightlifting belt."** Not only will this stiffen your spine and keep it safe when performing heavy lifts (like a squat or deadlift), but it will also help transfer force throughout your body. For example, a weightlifter performing a jerk requires sufficient core stability to transfer the power he or she generates from their legs through the core and into the upward drive of the barbell.

All for One and One for All

Much like the symphony orchestra analogy from before, every muscle of the core has a role to play, but none is more important than the others. For this reason, proper stability training should not focus on one specific muscle. For decades, medical practitioners were incorrectly taught to focus on and isolate certain muscles such as the transverse abdominus (TA), multifidus, or QL to enhance core stability. This method is flawed for a number of reasons.

First, research has shown that it is impossible for an individual to activate only one specific muscle of the core. Despite what your physical therapist or doctor says, you cannot train your multifidus, your QL, or even your TA muscle in isolation.

TRANSVERSE ABDOMINALS



The Big Three

Now that you know which types of exercises are superior for rehabbing most back injuries, it's time to discuss which exact exercises to start with. There is unfortunately no one-size-fits-all approach when it comes to core exercises because there is no one universal movement that equally stresses all of the muscles that surround the spine. For this reason, we must utilize a battery of exercises to work all of them efficiently.

Technically, *any* exercise can be a "core" exercise if it is performed with sufficient attention to creating and maintaining a stable spine. It all depends on the objectives of the exercise and the individual's needs. One key to fixing injured backs is to use exercises that enhance stability but place minimal stress on the spine while the exercises are being performed.

In his years of studying the spine, Dr. McGill has found three exercises that efficiently address all of these areas without placing excessive stress on the parts of the back that may be aggravated or irritated due to injury. This group of exercises has become known as "the Big Three":

- Curl-up
- Side plank
- Bird-dog

Mobility First

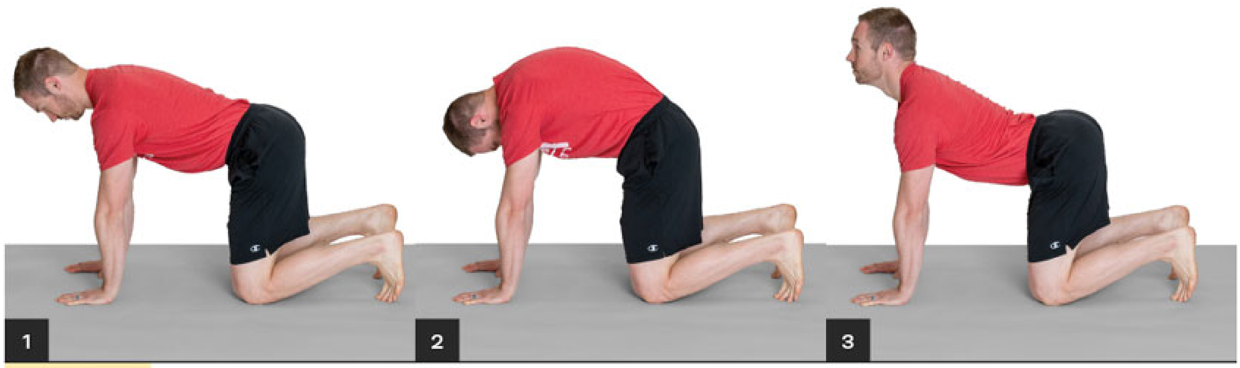
Before you begin core stability training, I recommend addressing mobility restrictions at the hip and/or thoracic spine.

If your mobility in either of these areas is limited, it can lead to movement compensations at the low back. For example, if there is limited hip mobility during the squat motion, the pelvis can be pulled underneath (posterior pelvis tilt), causing the lower back to leave its neutral position and round.

For this reason, if you perform only core stability work but don't address any significant mobility restrictions in joints above or below the lumbar spine, the stiffness you created will always be compromised.

After addressing mobility restrictions in those areas, Dr. McGill recommends that you perform the cat-camel before the Big Three to reduce low back stiffness and improve motion of the spine. Unlike other stretches for the low back that can place harmful stresses on the spine, this exercise emphasizes mobility in a spine-friendly manner.

Assume an all-fours (quadruped) position. Slowly arch your entire spine and hips into a flexed or rounded position as high as possible without pain. You should end with your head looking down toward the ground. This is the camel position. After pausing for a few seconds, move into the opposite downward extended position with your head looking up (the cat). Make sure you move into a *light* stretch for each position; do not force your spine into any pain.



CAT-CAMEL

Perform five or six cycles of this exercise before moving on to the Big Three, starting with the curl-up. [68]

Curl-Up

When most people perform what they have been taught to be a curl-up, they bend or flex their entire spine and attempt to bring their chest toward their knees. While this exercise does activate the anterior core muscles to a great degree (especially the rectus abdominis, or six-pack muscle), the "crunch" motion does a few things that aren't so appealing, especially for those dealing with back pain.

First, the motion of the classic curl-up places a large amount of compression on the spine that can flare up symptoms for those who are "load intolerant." [69]

Second, the motion pulls the spine out of its neutral slightly arched position and flattens it into a bit of flexion. If your low back symptoms increase when you bend your spine ("flexion intolerance"), you should avoid this motion for the time being.

The traditional curl-up also relies heavily on the psoas muscle of the anterior hip to pull the torso toward the thighs. So, while you think you may be isolating and sculpting that sexy six-pack by doing endless crunches, you're actually doing a really good job of strengthening your hip flexors. [70]

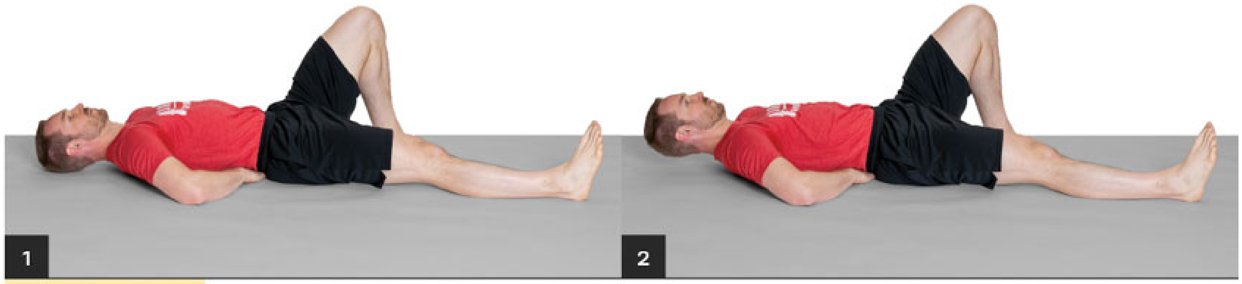
We can home in and focus our attention on improving the stabilizing ability of the anterior core muscles in a more efficient way by performing a modified curl-up.

Step 1: Lie on your back with one knee bent and the other straight. If you have pain that radiates down one leg, straighten that leg against the ground. Place your hands under your low back (this will ensure your spine remains in a neutral slightly arched position during the next step).

Step 2: Lift your head only 1 inch off the ground and hold that position for 10 seconds. If you're resting your head on a pillow, imagine it as a scale and lift your head enough to make the dial read zero. [71] The goal is to perform the curl-up without any movement in your low back! If you raise your head and shoulders too high (like a traditional curl-up or crunch), your low back will round, and excessive forces will be transferred to the spine that could increase your symptoms.

Step 3: After a 10-second hold, relax your head back down to the resting position.

You can progress this exercise by bracing your abs before moving your head or raising your elbows from the ground to decrease your base of stability. [72]



MCGILL CURL-UP

[73] This can be freely modified to suit your current level of endurance and goals: for instance, using a 6-4-2 or 8-6-4 rep scheme.

During these 10-second holds, focus on your breathing. Inhale and exhale slowly (about 5 times in a row for each repetition). As you breathe, concentrate on maintaining the contraction of your abdominal muscles. The cue to take "small sips of air in and out" (as if you are breathing through a straw) will teach you how to create sufficient core stability while moving throughout your day. You wouldn't hold your breath when climbing stairs or while performing a 10-repetition set of squats! Learning how to breathe and maintain sufficient core stability for the task you're going to carry out is a vital step in the early rehab process for low back pain.

The McGill Big Three has been highly effective since I started using it with my patients who come in for low back physical therapy. I have also found the Big Three beneficial in my own training as a weightlifter. I perform these core stability exercises before all my lifts.

Side Plank

Now that we've addressed the anterior core muscles, let's move to the sides of the body. The side plank is a unique exercise because it activates the lateral oblique and QL muscles on only one side of the body, making it an excellent choice for addressing weak links in stability while placing minimal force on the spine. It also engages an important stabilizer of the hip/pelvis on the lateral hip: the glute medius. [74]

SIDE PLANK

> **Step 1:** Lie on your side with your knees bent and your upper body supported through your elbow. Place your free hand on your opposite hip or shoulder.
>
> **Step 2:** Raise your hips so that only your knee and arm support your body weight. As you lift your body from the ground, push your hips forward with the cue "squat your hips up."

Step 3: Hold this position for 10 seconds before returning back down. Push your hips back using the cue "squat your hips down." Perform the same descending pyramid rep scheme for each side.

SQUAT HIPS DOWN

SQUAT HIPS UP

You can do a number of progressions with this exercise. You can start by moving your hand placement from your opposite shoulder to the top of your hip or even move to a full side plank, with your body weight supported by your feet and elbow.

FULL SIDE PLANK

SIDE-LYING LEG LIFT

the low back remains stable allows this exercise to have excellent carryover to movements you perform throughout the day and in the weight room.

Step 1: Assume an all-fours (quadruped) position with your back in a neutral alignment. Remember, a neutral position is slightly arched, not completely flat.

Step 2: Without allowing any movement to occur at the low back, kick one leg backward while simultaneously raising the opposite-side arm until both limbs are fully straightened. A helpful cue to make sure the leg movement doesn't create an overarching of your back is to think about kicking the heel of your foot straight back. Making a fist and contracting your arm muscles as you hold it in the extended position can also increase the muscle activity of your core, especially of your erector spinae muscles.

If you are unable to perform simultaneous arm and leg movement without pain, or if you find that you're losing balance, try the modified version with only leg movement.

Step 3: Hold each extended pose for 10 seconds before returning to the all-fours position. You can also "sweep" your arm and leg back underneath your body between repetitions. Don't let your back round during this motion; instead, maintain a neutral spine and allow motion to occur only from the hips and shoulders! Perform the same descending rep scheme as in the previous two exercises. You can progress this exercise by drawing a square with only your outstretched hand or both your hand and opposite foot.

BIRD DOG QUADRUPED

EXTENDED BIRD DOG

[75] Always start this exercise by stiffening your core *before* moving your limbs. Focus on your breathing and on maintaining sufficient abdominal contraction during your 10-second pose *and* during the “sweep” motion in between.

SWEEP UNDER & TOUCH

What About Back Stretching?

If you didn’t notice, the prior exercises to kick-start the rehabilitation process do not include any stretches for the back! Early in my career as a physical therapist, it was common to prescribe certain stretches for back pain, like pulling the knees to the chest while lying on your back.

KNEES TO CHEST STRETCH

At the time, this exercise made sense. Those who had difficulty standing for long periods or lying flat on their back often felt better in a flexed position. Many who complained of low back stiffness and pain found instant relief of their symptoms after performing a few of these stretches.

However, after reading and studying with Dr. McGill, I realized that for most people, this relief is only temporary. Stretching the low back stimulates the stretch receptors deep inside the muscles, giving the perception of pain relief and the feeling of less stiffness.

Most of the muscle pain and stiffness you feel in your back is a consequence of a chemical reaction called *inflammation* that occurs from the real injury located deeper in the spine (disc bulge, facet irritation, etc.).[76] This underlying injury is what causes the secondary contraction or spasm of the surrounding muscles.

For this reason, a large majority of athletes should aim to stabilize the core and restore proficient movement in order to successfully rehab a back injury. Stretching the low back only treats the symptoms and does not address the true cause of the pain.

Reawaken Those Sleeping Glutes!

It is common to see athletes with back pain also have an inability to properly activate and coordinate their glute muscles. Simply put, the butt muscles can fall asleep. [77] When this happens, the body naturally starts to use the hamstrings and low back muscles more to create hip extension; both are problematic in creating efficient movement and place excessive stress on the spine. [78]

If the single-leg bridge test (see here) exposed a problem in how your body coordinates and turns on the glutes, the following exercises should help.

Bridge

Step 1: Lie on your back with your knees bent.

Step 2: Squeeze your butt muscles *first* and *then* lift your hips from the ground. Squeeze your glutes as hard as you can for 5 seconds before coming back down. Eventually, work your way up to a 10-second hold.

If your hamstrings start cramping, two modifications can help. First, bring your heels closer to your hips. Doing so shortens the length of the hamstrings and puts them at a disadvantage to contribute to the movement (a concept called *active insufficiency*). [79] You can also jam your toes into the ground and think about pushing your feet away from your hips. This will turn on your quads slightly, which in turn will decrease hamstring activity (a concept called *reciprocal inhibition*). The only muscles left to create hip extension with the hamstrings out of the picture are—you guessed it—the glutes!

Recommended sets/reps: 2 sets of 20 reps for a 5-second hold

DOUBLE-LEG BRIDGE

If you still notice back pain when performing the bridge with the prior cues, try this modification. Position yourself with your head near a wall and push your hands into the wall prior to performing the bridge. Pushing into the wall engages your core, thereby increasing the efficiency of the bridge and hopefully decreasing your pain.

DOUBLE-LEG BRIDGE WITH HANDS INTO WALL

DEEP SQUAT WITH PLATE STRETCH

ISOMETRIC SQUAT HOLD

[80]

The combination of these exercises should be safely performed daily if you are trying to recover from back injury but should not be performed directly after rising from bed in the morning, when the discs of your spine are the most hydrated and most prone to injury. [81]

The Early Rehab Plan

Now that you know your specific triggers for pain and have a good understanding of the McGill Big Three exercises, you can start to piece together a well-organized rehab program to follow.

Dr. McGill recommends coupling a regimented walking program with the Big Three exercises every day. [82] Getting up and walking throughout the day can be extremely helpful for maintaining the health of your spine and your base level of fitness after a small setback of a back injury. Start with smaller bouts of walking—5 to 10 minutes with a fast pace that causes you to swing your arms. The goal should be to eventually reach a 10- to 15-minute walk 3 times a day.

Do the Big Three every day for the first few weeks. You can then progress the frequency to twice per day while keeping your overall volume the same. For example, start by performing the 6-4-2 descending rep scheme once per day. Then progress to performing a 3-2-1 descending rep scheme of each exercise held for 10 seconds twice per day.

Bridging the Gap: Early Rehab to Performance

Resolving any kind of pain requires a three-stage approach. You must first eliminate the cause of the pain to decrease your symptoms and provide an optimal environment to kick-start your body's natural healing process. This means avoiding your individual triggers for symptoms at all costs! While this step seems easy, many athletes have a hard time complying with it. When I advise an athlete to stop squatting or deadlifting for the time being, I'm often met with the line, "I can't stop completely or I'll lose the gains I've made!"

Have some perspective and drop your ego. You will *not* lose all that you have gained by taking a few steps back in the short term. No rehabilitation plan or corrective exercise program will truly fix your pain and restore your body in the long term if you are too stubborn to deviate from a training plan that is creating pain. If certain lifts cause you pain, stop performing them for now. Have faith that you can and will return to them eventually.

The second stage of rehab is to address your body's weak links that led to pain in the first place. This includes a well-rounded approach of starting back-friendly core stability work like the McGill Big Three, doing mobility/flexibility exercises for restricted joints, and retraining poor movement patterns.

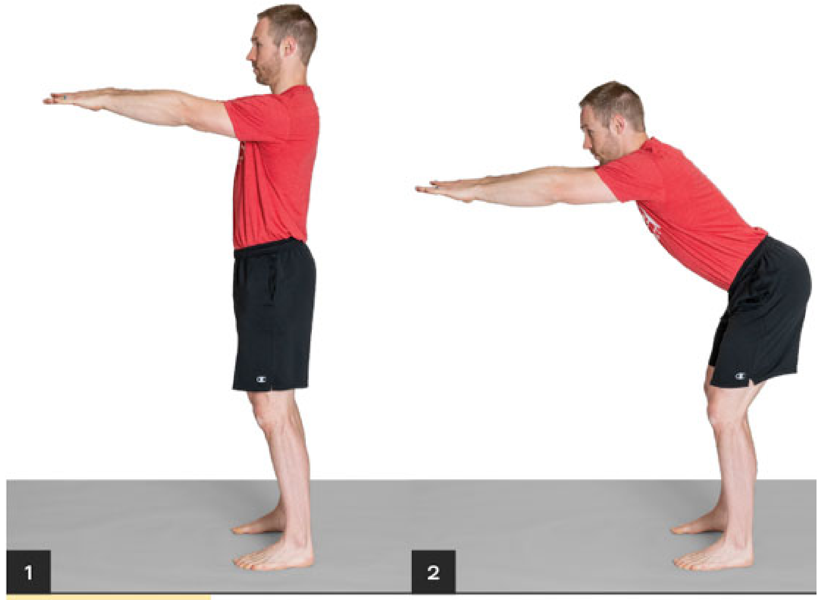
Learning the Hip Hinge

One of the most common reasons for developing back pain is an inability to use the hips properly. [83] From bending over to pick up a pencil off the ground to grabbing a loaded barbell, many who develop back pain have forgotten how to move about the hips and instead have allowed the lumbar spine to become unstable and move excessively. Therefore, learning to move from the hips and keep the spine stable (a movement called a *hip hinge*) is an essential part of the rebuilding process. This step is crucial for those who are "flexion intolerant."

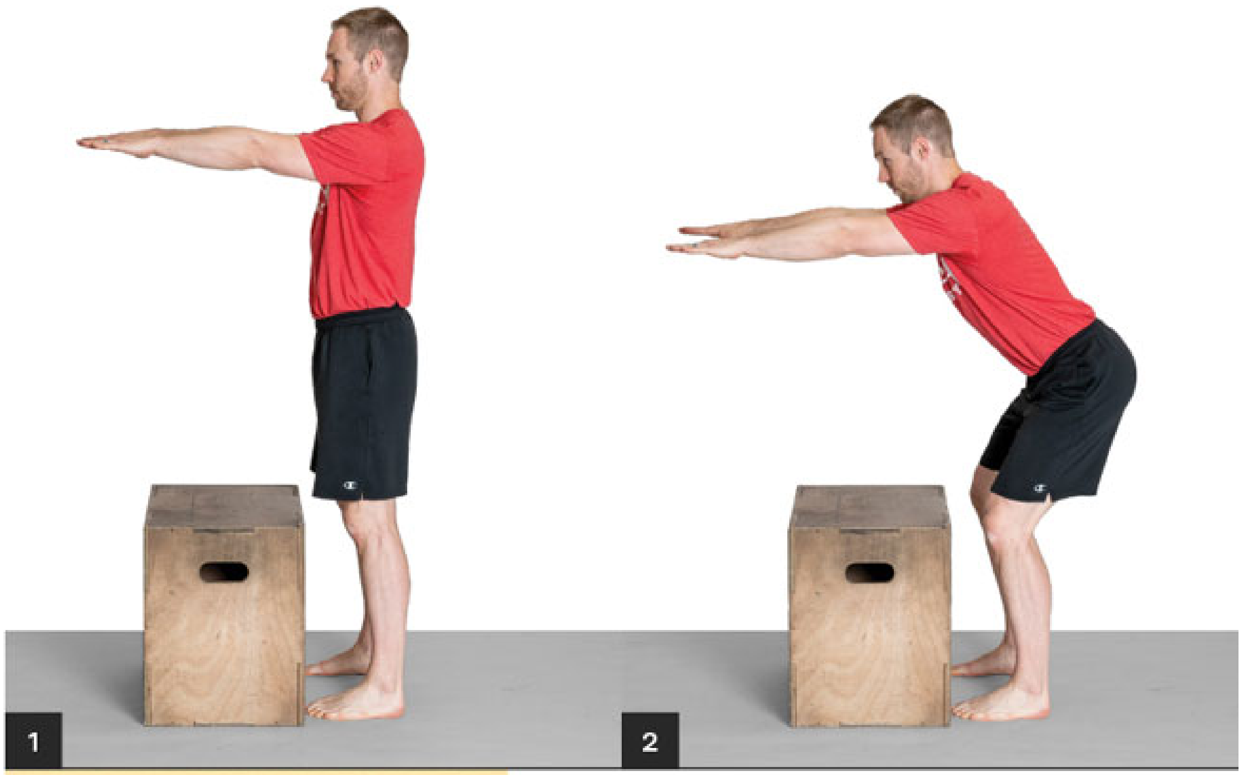
To perform a proper hip hinge, stand with your hands straight out in front of you. Grip the ground with your feet (big toes into the ground) and feel for your body weight to be spread evenly across your entire foot. Next, drive your knees out to the sides to engage your lateral glutes, making sure to keep your feet firmly glued to the ground. Looping a small resistance band around your knees early on in the learning process can be a great way to teach your body how to create sufficient tension in your lateral hip muscles.

Start by pushing your butt backward and bringing your chest forward. (No motion should occur at your back.) Push your arms forward so that they're parallel to the ground; this will help counterbalance your hips going backward. Without allowing your knees to move forward, squat down a few inches and hold this position for a few seconds. If you did it correctly, you should feel tension building in your glutes and hamstrings.

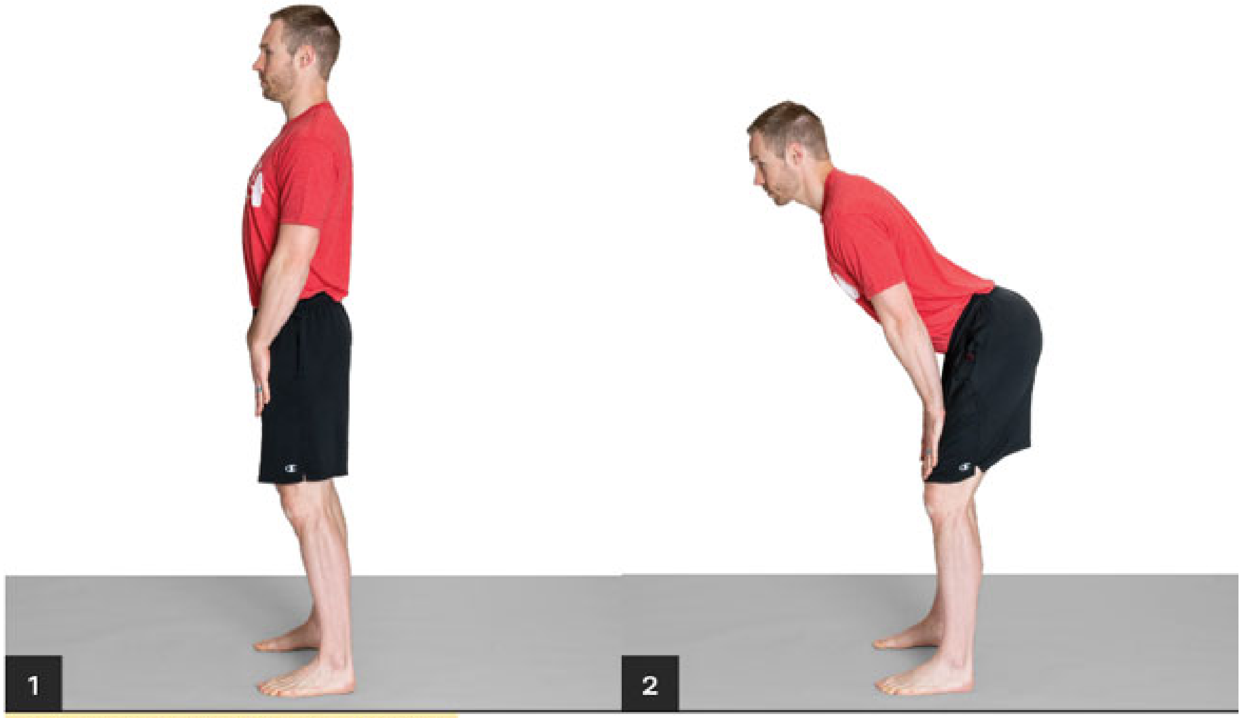
STANDING HINGE START
(ANGLE VIEW)



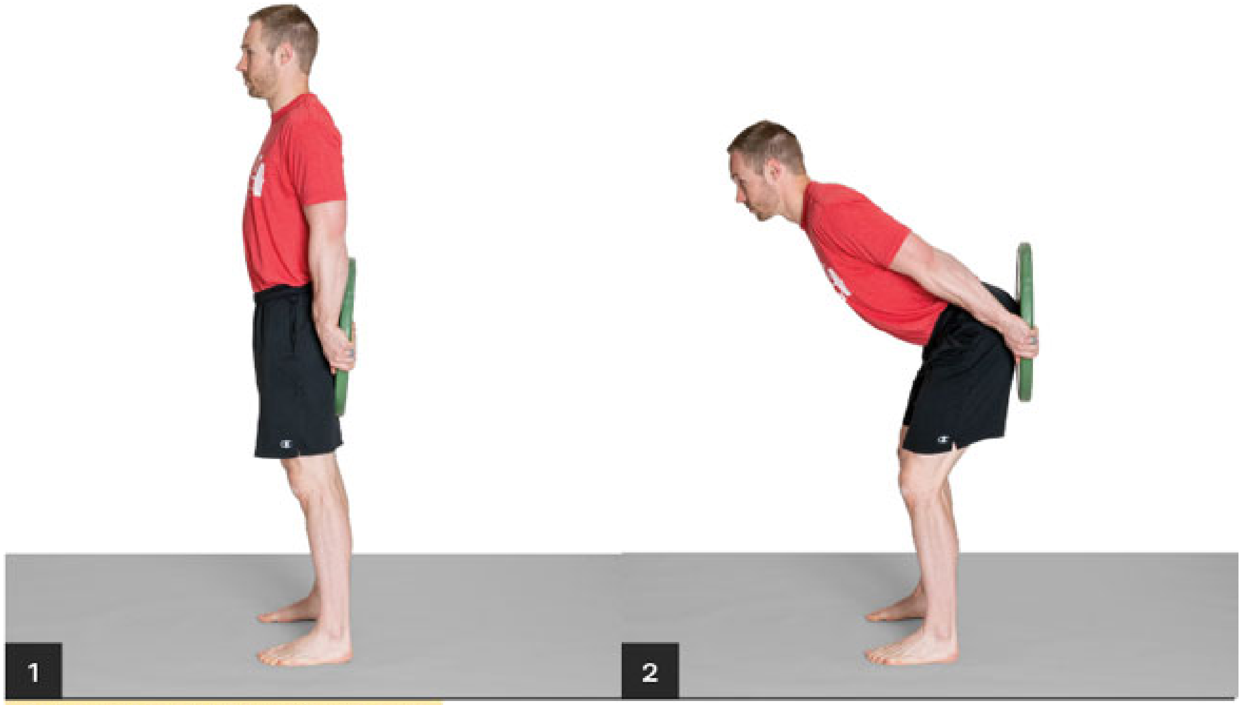
STANDING HINGE
(SIDE VIEW)



HINGE WITH BOX IN FRONT OF TOES



HINGE WITH HANDS ON THIGHS



HINGE WITH PLATE ON GLUTES

core stability through multiple planes of motion and movement patterns. The exercises in this third stage of rehab fall into four categories:

- **Push:** squat, sled push
- **Pull:** deadlift, inverted row
- **Carry:** suitcase carry
- **Anti-rotation:** Pallof press, one-arm row

Drawing from each of these categories will help you rebuild your body by forcing you to expose and address "weak links" that often go unnoticed in performance training. For example, a powerlifter will often perform exercises during training that fall into the push, pull, and anti-rotation categories (with exercises like the squat, deadlift, and bench press), but if they never perform any loaded carry exercises, they essentially leave their body open to potential injury because they fail to stimulate core stability in the frontal (forward and backward) plane of motion.

When returning from a back injury, you must take a careful approach during this third stage of rehab and choose exercises that facilitate healing without placing excessive load on your spine that re-creates your symptoms. For example, performing anti-rotation exercises before you can successfully squat with light weight often spells disaster. Exercises that create a twisting force on the body, like the Pallof press, can place upwards of four times as much compression on the spine compared to the same weight that attempts to create a flexion/extension force. [84]

The following is a logical progression of stability exercises that stress the body first through the sagittal plane (flexion/extension torque), then the frontal plane (lateral torque), and finally through the transverse plane (torsion torque). While there is no such thing as an ideal set of exercises for any rehabilitation program, these exercises can be a springboard for creating the plan that best suits your body and your goals.

As you work through each exercise, be cautious of how quickly you increase load. An efficient rehab program slowly applies load to the body. Too little and the body fails to adapt and remains fragile; too much and we overstep the adaptation process and pain returns. Listen to your body and remove your ego from the equation.

Squat

Once you can perform a hip hinge without pain, it's time to start rebuilding the squat. Start by performing bodyweight squats, increasing the depth as you are able to do so without pain. Depending on the severity of your injury, it may be a few weeks before you are able to regain a full-depth bodyweight squat without symptoms. When you can, it is time to load the movement.

The position of the load (on your shoulders versus on your back) and the amount of weight being lifted will dictate how much force is placed on your spine. For example, performing a goblet squat with a 30-pound kettlebell held by your chest places less force on your spine than a 135-pound front squat. In the same manner, a 135-pound front squat creates less torque on the lumbar spine compared to the same weight in a back squat. The more vertical torso assumed during the front squat to stay balanced provides a smaller moment arm (the distance from the vertical line of gravity pulling down on the barbell to the joints of the lumbar spine) than the relatively inclined torso of the back squat. [85]


GOBLET SQUAT


FRONT SQUAT


BACK SQUAT

trunk will instantly become more stable and capable of handling tremendous weight. This is how you activate your body's natural "weightlifting belt."

To keep this increased pressure in your abdomen and your spinal stability high, you must forcefully stop your exhale from escaping; this is known as the Valsalva maneuver. By grunting or using a *tss* sound as you slowly exhale through a small hole in your lips, you can maintain a sufficient amount of stability in your core and keep your spine safe. [86]

This held breath should not be used for more than a few seconds during any lift, as doing so could increase your blood pressure to harmful levels and cause blackouts. If you have a history of cardiovascular issues, speak with your doctor before using this technique. For healthy athletes, however, a small temporary rise in blood pressure is not harmful.

Sled Push

Successfully pushing a weighted sled requires you to generate tremendous force with your legs and transfer it through a stiffened core into your arms and eventually the sled. If you are unable to create sufficient stability, your spine will move out of the ideal neutral position, causing energy to "leak out."

Start with your hands high on the sled poles. Before making any movement, grip the poles as hard as you can and drive your feet into the ground. Along with bracing your core, these actions will help create the spinal stiffness necessary to drive the sled with incredible power while keeping your back safe.

SLED PUSH

DEADLIFT START WITH MOMENT ARMS

DEADLIFT FROM BLOCKS

squeezing your arms tight to your rib cage. This action engages your powerful lat muscles and creates a tremendous amount of core stiffness and upper body stability.

Next, engage the rest of your upper body by pulling upward against the bar—not moving it from the raised platform yet, but enough to essentially wedge your body against the bar like a "trust fall." If you're doing it correctly, you'll see the bar bend upward slightly—an action some call "taking the slack out of the bar."

TAKING THE SLACK OUT OF THE BAR

Micah Mariano deadlift

While you're creating this tension with your upper body, you need to simultaneously do the opposite with your lower body. This means creating a tremendous amount of tension in your glutes and hamstrings as you drive your feet hard into the ground. Finding the right balance between pulling on the bar and pushing through the floor will help keep your back safe. As you progress with this lift, you can manipulate the different variables by adding weight or decreasing the height of the blocks.

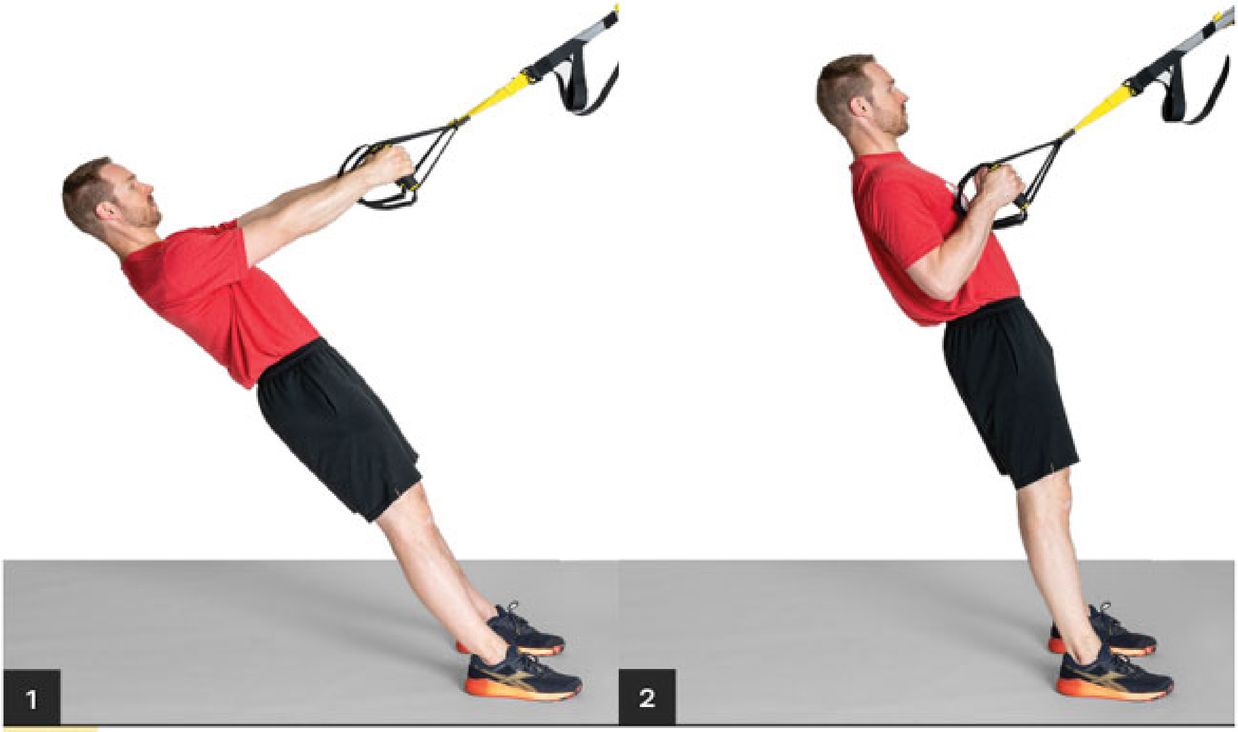
Modifying your deadlift technique can also be helpful during the short-term rehabilitation process. For example, a sumo-style deadlift is often more friendly to the low back compared to the torso angle used during a conventional deadlift. The wider stance of the sumo deadlift allows you to keep the bar closer to your body and maintain a more upright torso position. Both of these factors shorten the moment arm on the lumbar spine and result in less overall load. [87]

SUMO DEADLIFT

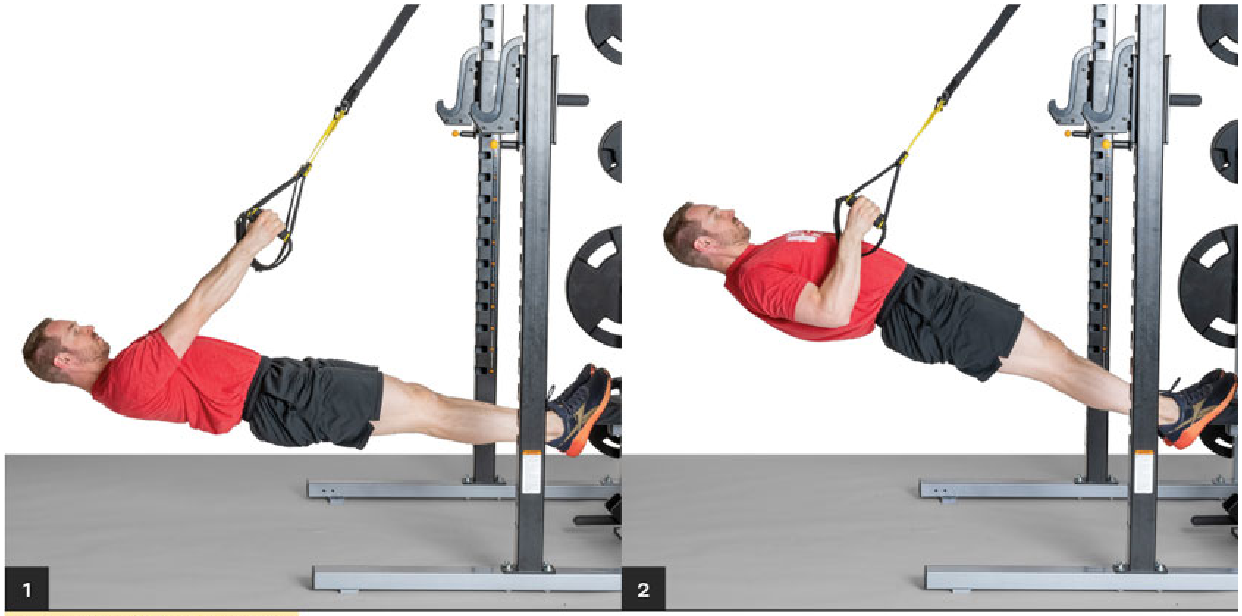
CONVENTIONAL DEADLIFT

Inverted Row

The inverted row, performed with a suspension trainer or gymnastic rings, has been shown in research to elicit a high amount of upper and mid-back muscle activation while placing minimal stress on the spine. [89] This makes this variation of the row a great exercise for the early stages of rehabilitation from a back injury. Make sure to focus on your breathing when performing this exercise. Take a big breath and brace your core, and *then* initiate the rowing movement.



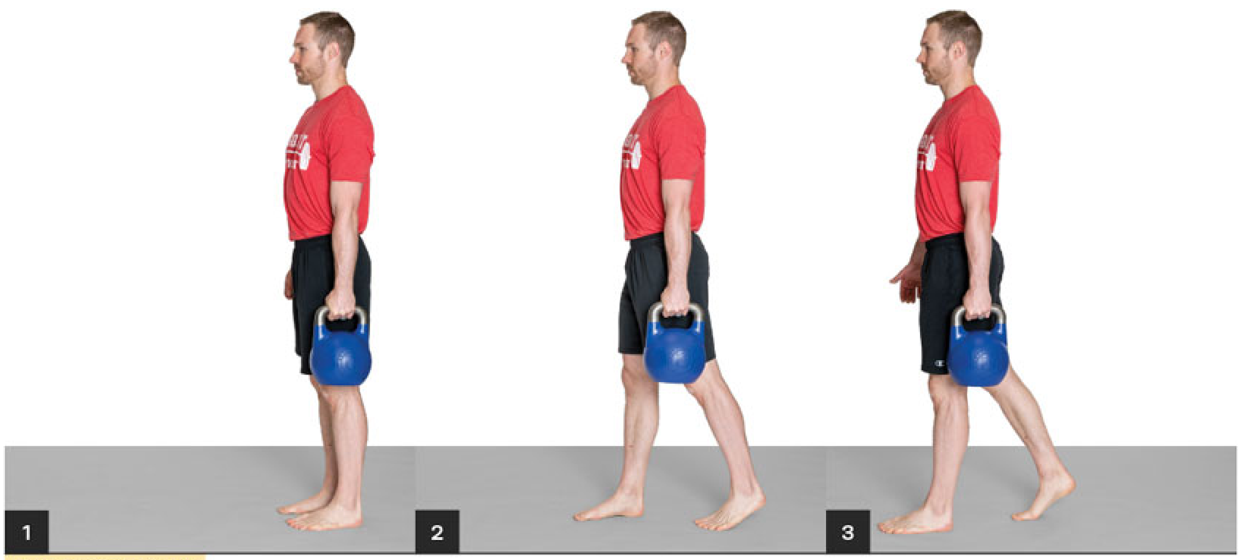
ROW



ROW WITH FEET ELEVATED

backward an athlete takes moves their body through the frontal plane of motion.

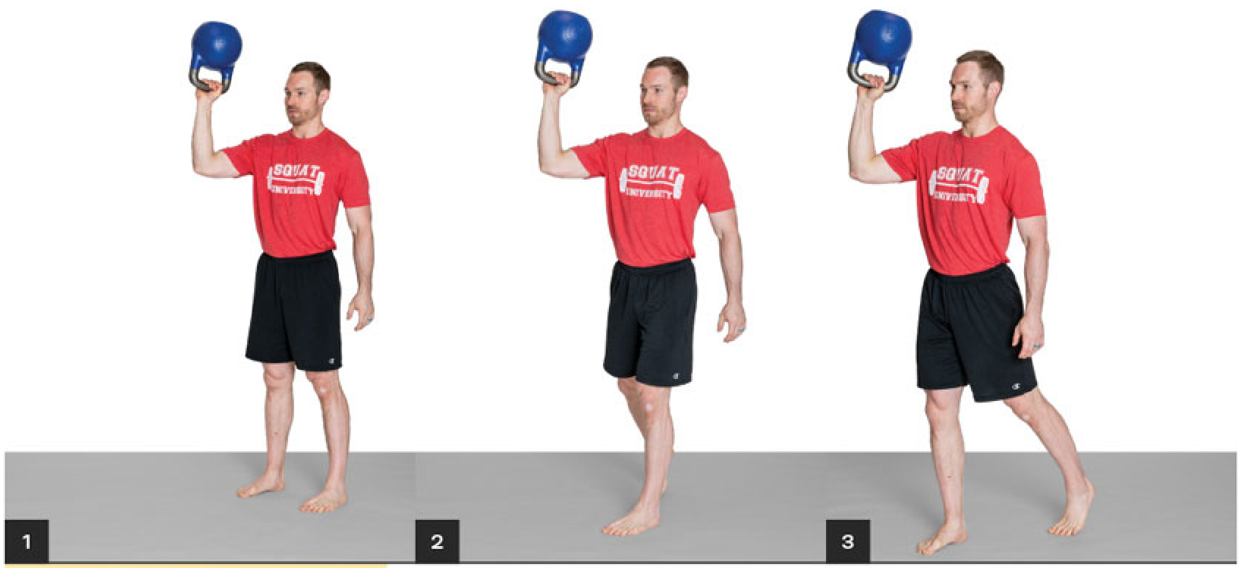
The suitcase carry challenges the core in the frontal plane and will help address this common imbalance for many athletes. To perform it, grab a lightweight kettlebell or dumbbell (10 to 20 pounds to start) with one hand and hold it at your side. Brace your core and squeeze your arm to your side to activate your lats; use the cue “make your armpits disappear” to create upper body tension when setting up. Grip the kettlebell handle as tightly as you can before you start. As you walk, concentrate on limiting side-to-side leaning of your torso.



SUITCASE CARRY

Performing a suitcase carry with the weight in one hand is significantly harder and poses a greater challenge to your core than performing the exercise with weight in both hands. In fact, research has shown that carrying double the weight in each hand (using two 30-kilogram/66-pound kettlebells) places significantly less compression on your spine than carrying a single 30-kilogram/66-pound kettlebell in one hand. [90] This is because the effort to stabilize the uneven forces moving up your spine by holding the weight with one arm challenges your body to a greater degree.

You can increase the stability challenge to your core by holding the kettlebell in the upside-down position. Brace your core and *then* start your walk. If you want to progress this exercise one step further, lift your thighs and march in a slow, controlled manner.



UPSIDE-DOWN KETTLEBELL CARRY

The upside-down kettlebell carry is one of the most challenging variations. If you struggle to keep the kettlebell from wobbling, you may want to blame your grip strength. However, it is often a lack of sufficient core stability that leads to the weight falling over while you walk.

Research has shown that back pain diminishes our sense of spinal position and postural control. [91] The more stable you can keep your core, the easier it will be to keep the kettlebell from wavering. Therefore, instead of focusing on the kettlebell, focus your attention on maintaining a braced core.

Pallof Press

The lateral banded press or Pallof press challenges the body's ability to resist a twisting motion. This provides a unique test for strength athletes who rarely encounter twisting forces during normal barbell training.

Begin in a standing position with your hands close to your stomach and holding a band/cable. The attachment for the band should be perpendicular to your side. Brace your core and push your hands away from your body. Hold this position for five seconds without allowing your body to twist.

PALLOF PRESS

You can alter this exercise by assuming different positions, such as half-kneeling, tall kneeling, or even a split stance. You could also add a lift after the initial press and hold. Another variation is squatting while doing the press. You can play with different squat depths to challenge your core and hips at different angles.

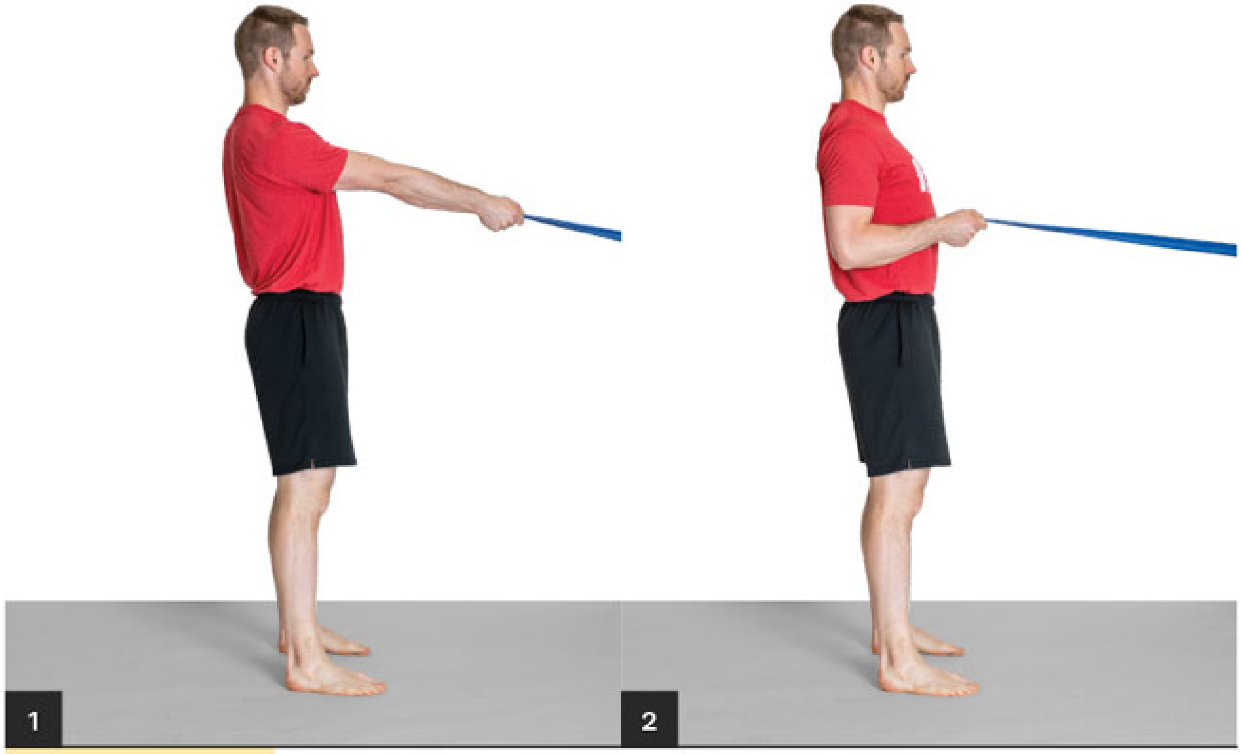
Recommended sets/reps: 2 or 3 sets of 10 reps on each side


PALLOF PRESS SPLIT STANCE


PALLOF PRESS IN DEEP SQUAT

Stand facing a cable machine or attach a band around a rig. Perform a row with one arm while maintaining a neutral spine. Don't allow your body to twist at all. After a two-second hold, extend your arm back to the start position. (Turn the page for photos of this exercise.)

You can perform this row in a number of positions, such as in a lunge or kneeling.



ONE-ARMED ROW

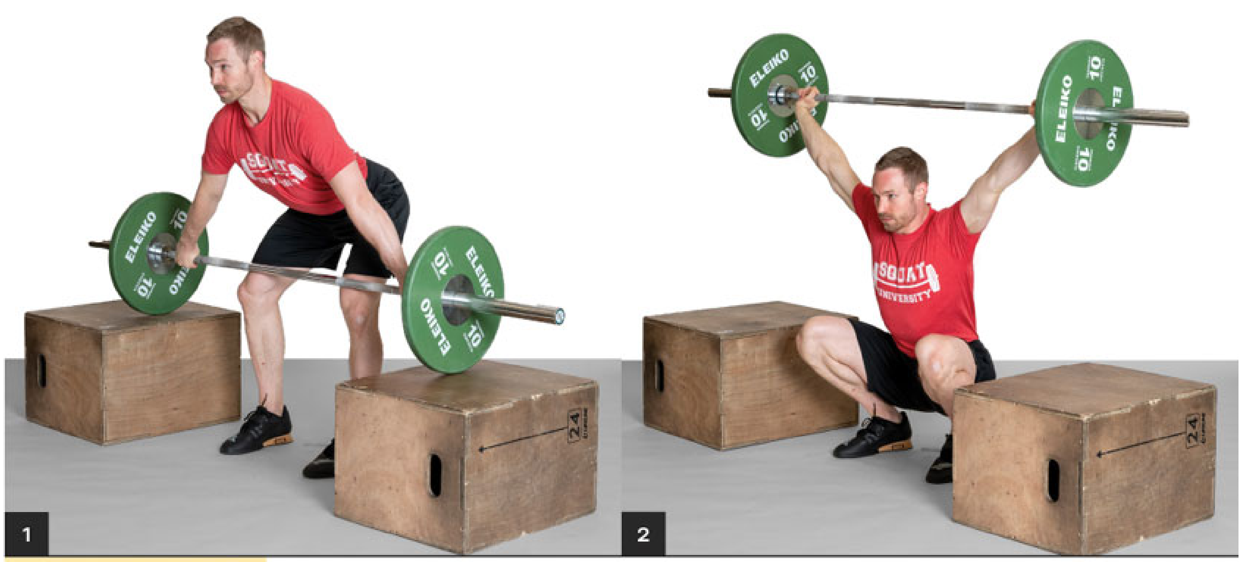
following exercise modifications in your training.

Olympic Lifts

The Olympic lifts (the snatch and the clean and jerk) can eventually be added to the rehabilitation program. Due to their complexity and speed requirements, however, I do not recommend reintegrating these lifts into your program until you can easily perform a single repetition of both the squat and the deadlift at 70 percent of your prior 1-rep max without pain.

As with the deadlift, using blocks initially can help you to perform either Olympic lift with less load on your low back. Start with a block height that positions the barbell at mid- to low thigh (above the knee). Research has shown that assuming a more vertical trunk position as the barbell passes the knees places significantly less stress on the low back during either the snatch or the clean and jerk (meaning that the start of the full lift from the ground is the most demanding on the spine).[88]

As you progress this lift, you can manipulate the different variables by adding weight or decreasing the block height.

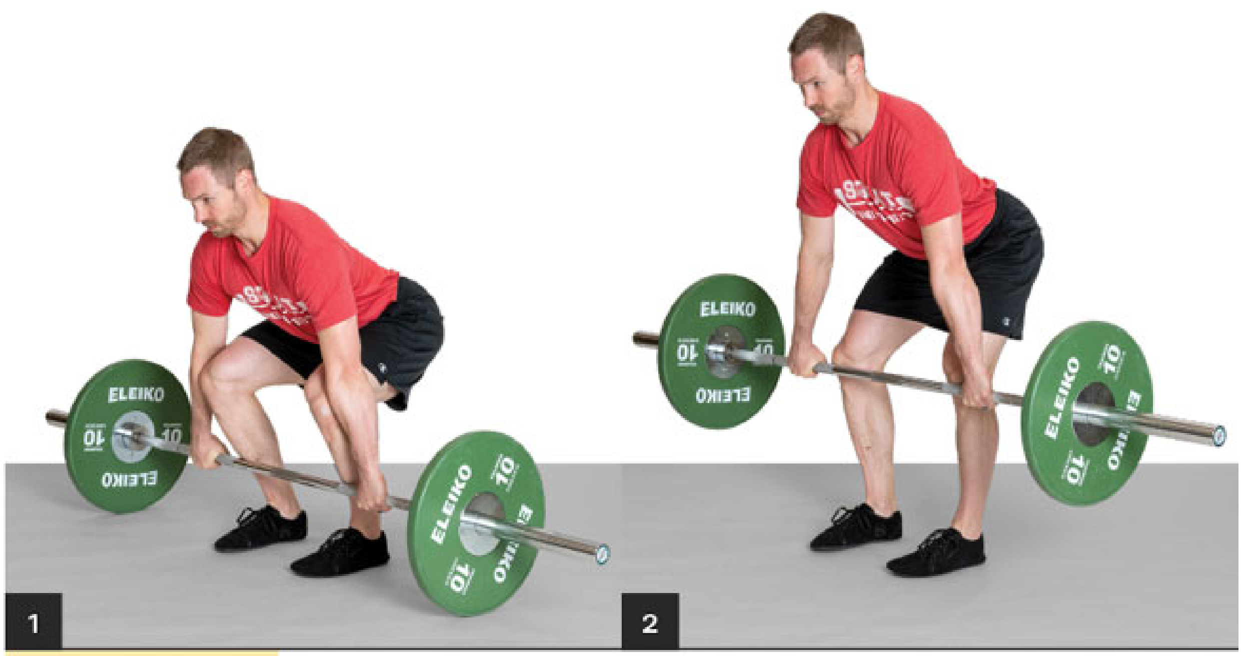


SNATCH FROM BLOCKS

Paused Deadlift

The paused deadlift is a great variation to reinforce proper pulling technique. To perform it, start with the exact same approach you would use for a regular deadlift. Take a huge breath, brace your core, and slowly pull the bar from the ground. You can perform two- to five-second pauses in a variety of positions, such as mid-shin, below the knee, and above the knee. I also like to add a pause at the knee on the way back down for an added challenge.

During the pause, feel your legs pushing hard into the floor. If you maintain sufficient core stability by holding your breath and bracing your core, you should not feel any recurring pain in your low back. Perform up to three reps per set.



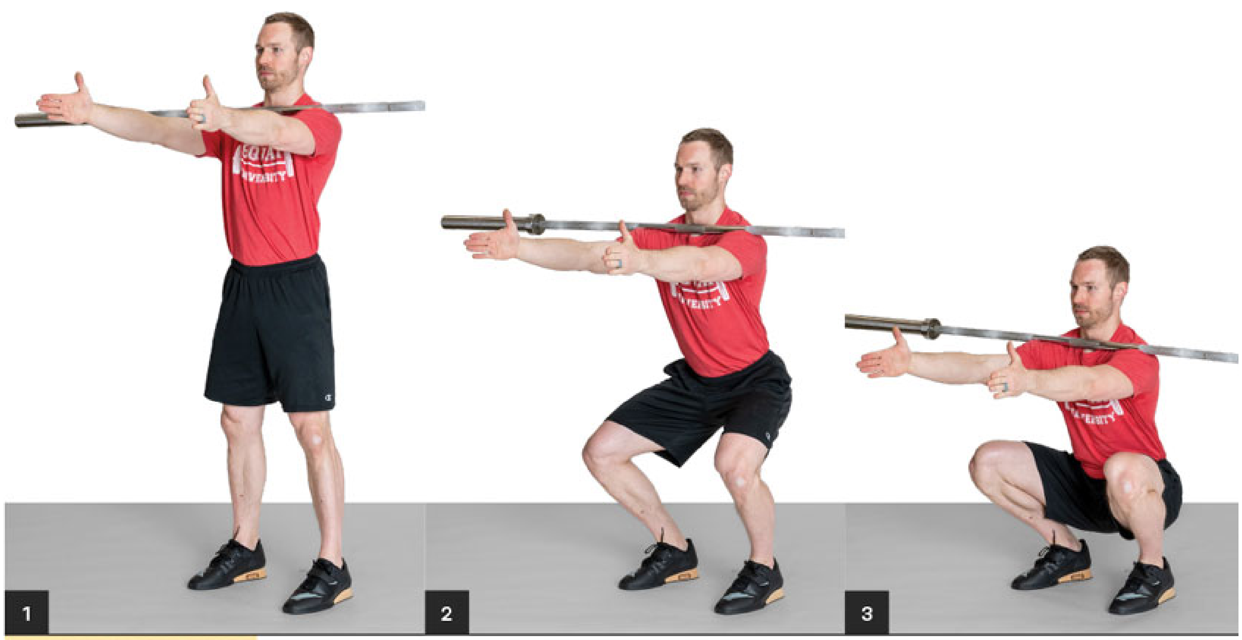
PAUSED DEADLIFT

Zombie Front Squat

When performing a front squat or a clean, you want to maintain as much of an upright trunk as possible. Athletes commonly hit a decent-looking bottom position with each rep but fail to maintain an upright torso on the ascent, allowing their backs to excessively round into flexion. The no-hands zombie front squat solves this problem.

Start by holding a barbell on your chest as if you were performing a front squat. Extend your arms straight out in front of you so that the barbell is resting solely on the tops of your shoulders and chest. Set your feet in a stable position by grabbing the ground with your toes. Take a big breath, brace your core, and begin your squat.

To maintain the bar on your chest and keep it from rolling off and onto the ground, you must keep an upright torso! Start with sets of one to three reps with light weight and progress in weight only as you can continue to maintain perfect technique.



ZOMBIE FRONT SQUAT

Squatting with Chains

The use of chains has been a popular method in the powerlifting community for many decades since it was introduced by Louie Simmons of Westside Barbell. [93] The theory behind their use is to accommodate for how the body naturally responds to moving weights, called the *force velocity curve*.

For example, everyone can picture that one lifter who does quarter squats with a lot of weight on the bar but has zero chance of completing the same weight to full depth. This is because the body is able to generate much greater force at the top portion of the lift as compared to the bottom.

Let's say an athlete's one-rep max squat is 300 pounds. They could load a barbell to 220 pounds and hang two chains per side at 20 pounds each. The total weight at the top of the lift would be the lifter's max of 300 pounds. As they drop into the bottom of the squat, the chains hit the ground and pile up, which decreases the overall weight pulling down. Therefore, at the top of the lift, there are 300 pounds on the bar, but at the bottom there are only 220 pounds. During the ascent, the weight is then slowly reapplied to the barbell (this is called *accommodating resistance*).

With a regular barbell squat, the weakest position for most people is at the bottom. The closer you get to the top position, the easier the motion usually becomes. Using chains allows you to work with heavy weights at the top part of the squat and decreases your chance of failing a rep when you're coming out of the bottom. Refer to the next page for photos.